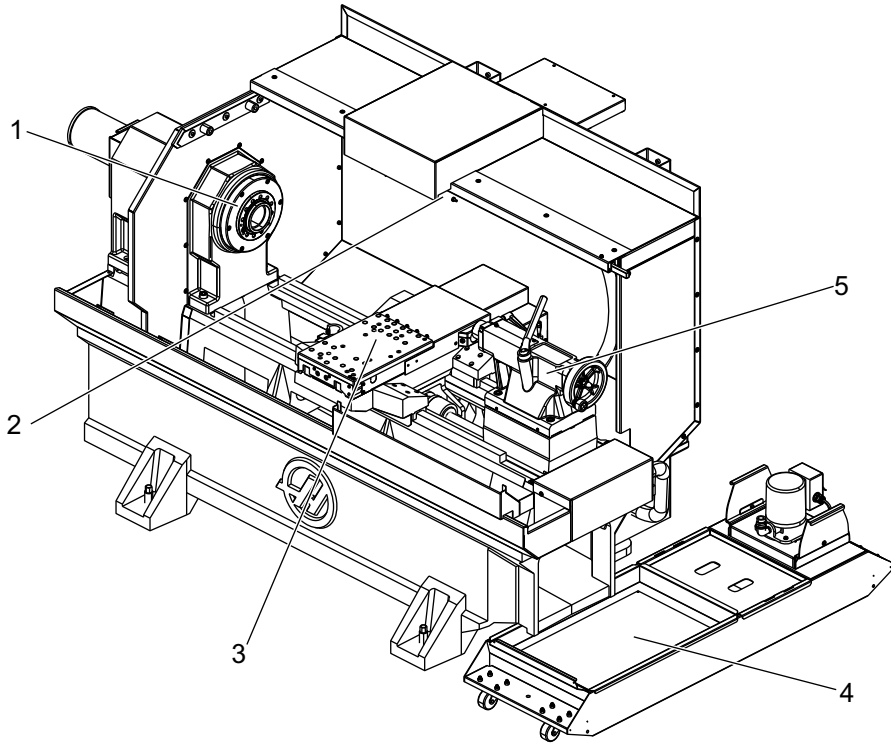


F2.7: Toolroom Lathe (front view, doors removed)



1. Spindle Nose
2. Worklight
3. Cross Slide (tool post / turret not shown)
4. Coolant Tank
5. Tailstock

2.2 Control Pendant

The control pendant is the main interface to your Haas machine. This is where you program and run your CNC machining projects. This control pendant orientation section describes the different pendant sections:

- Pendant front panel
- Pendant right side, top, and bottom
- Keyboard
- Control display

2.2.1 Pendant Front Panel

T2.1: Front Panel Controls

Name	Image	Function
[POWER ON]		Powers the machine on.
[POWER OFF]		Powers the machine off.
[EMERGENCY STOP]		Press to stop all axis motion, disable servos, stop the spindle and tool changer, and turn off the coolant pump.
[HANDLE JOG]		This is used to jog axes (select in [HANDLE JOG] Mode). Also used to scroll through program code or menu items while editing.
[CYCLE START]		Starts a program. This button is also used to start a program simulation in graphics mode.
[FEED HOLD]		Stops all axis motion during a program. The spindle continues to run. Press [CYCLE START] to cancel.

2.2.2 Pendant Right Side, Top, and Bottom Panels

The following tables describe the right side, top, and bottom of the pendant.

T2.2: Right Side Panel Controls

Name	Image	Function
USB		Plug compatible USB devices into this port. It has a removable dust cap.
Memory Lock		In the locked position, this keyswitch prevents alterations to programs, settings, parameters, and offsets.
Setup Mode		In the locked position, this keyswitch enables all machine safety features. Unlock allows setup (refer to “Setup Mode” in the Safety section of this manual for details).
Second Home		Press to rapid all axes to the coordinates specified in G154 P20 (if equipped).
Servo Auto Door Override		Press this button to open or close the Servo Auto Door (if equipped).
Worklight		These buttons toggle the internal worklight and High Intensity Lighting (if equipped).

T2.3: Pendant Top Panel

Beacon Light	
Provides quick visual confirmation of the machine's current status. There are five different beacon states:	
Light Status	Meaning
Off	The machine is idle.
Solid Green	The machine is running.

Beacon Light	
Flashing Green	The machine is stopped, but is in a ready state. Operator input is required to continue.
Flashing Red	A fault has occurred, or the machine is in Emergency Stop.
Flashing Yellow	A tool has expired, and the Tool Wear warning icon is shown.

T2.4: Pendant Bottom Panel

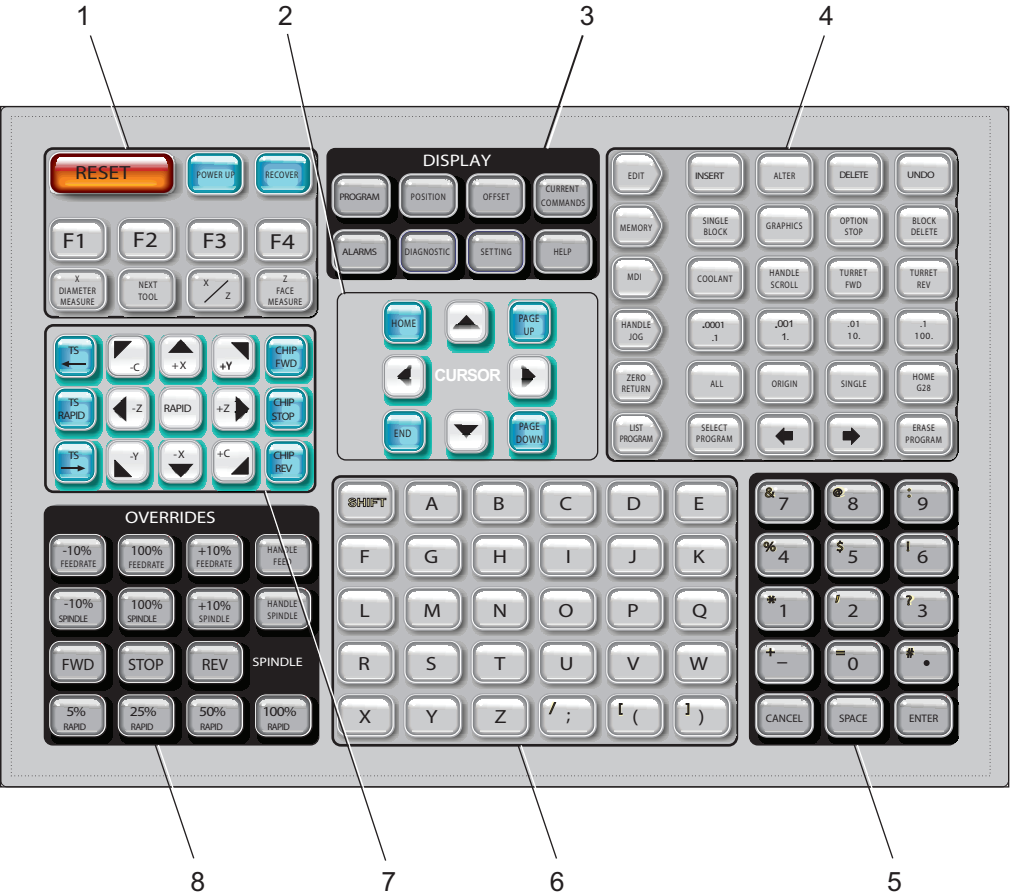
Name	Function
Keyboard Beeper	Located on the bottom of the control pendant. Turn the cover to adjust the volume.

2.2.3 Keyboard

Keyboard keys are grouped into these functional areas:

1. Function
2. Cursor
3. Display
4. Mode
5. Numeric
6. Alpha
7. Jog
8. Overrides

F2.8: Lathe Keyboard: Function Keys [1], Cursor Keys [2], Display Keys [3], Mode Keys [4], Numeric Keys [5], Alpha Keys [6], Jog Keys [7], Override Keys [8]



Function Keys

Name	Key	Function
Reset	[RESET]	Clears alarms. Sets overrides to default values.
Power up	[POWER UP]	Zero All Axes screen displays. Select the axis zero return order.

Name	Key	Function
Recover	[RECOVER]	The Tap Recovery screen displays. This button is functional to recover from a tap.
F1- F4	[F1 - F4]	These buttons have different functions depending on tab that is active.
X Diameter Measure	[X DIAMETER MEASURE]	Records X-Axis tool shift offsets on the offset page during part setup.
Next Tool	[NEXT TOOL]	Selects the next tool from the turret (usually used during part setup).
X/Z	[X/Z]	Toggles between X-axis and Z-Axis jog modes during part setup.
Z Face Measure	[Z FACE MEASURE]	Used to record Z-Axis tool shift offsets on the offset page during part setup.

Cursor Keys

The cursor keys let you move between data fields, scroll through programs, and navigate through tabbed menus.

T2.5: Cursor Key List

Name	Key	Function
Home	[HOME]	Moves the cursor to the top-most item on the screen; in editing, this is the top left block of the program.
Cursor Arrows	[UP], [DOWN], [LEFT], [RIGHT]	Moves one item, block, or field in the associated direction. The keys depict arrows, but this manual refers to these keys by their spelled-out names.
Page Up, Page Down	[PAGE UP] / [PAGE DOWN]	Used to change displays or move up/down one page when viewing a program.
End	[END]	Moves the cursor to the bottom-most item on the screen. In editing, this is the last block of the program.

Display Keys

You use the Display keys to see the machine displays, operational information, and help pages.

T2.6: List of Display Keys and How They Operate

Name	Key	Function
Program	[PROGRAM]	Selects the active program pane in most modes.
Position	[POSITION]	Selects the positions display.
Offsets	[OFFSET]	Displays the Tool Offset and Work Offset tabbed menu.
Current Commands	[CURRENT COMMANDS]	Displays menus for Devices, Timers, Macros, Active Codes, Calculators, Advanced Tool Management (ATM), Tool Table, and Media.
Alarms	[ALARMS]	Displays the Alarm viewer and Message screens.
Diagnostics	[DIAGNOSTIC]	Displays tabs for Features, Compensation, Diagnostics, and Maintenance.
Settings	[SETTING]	Displays and allows changing of user settings.
Help	[HELP]	Displays help information.

Mode Keys

Mode keys change the operational state of the machine. Each mode key is arrow shaped and points to the row of keys that perform functions related to that mode key. The current mode is always displayed in the top left of the screen, in *Mode : Key* display form.



NOTE:

[EDIT] and **[LIST PROGRAM]** can also act as display keys, where you can access program editors and the device manager without changing the machine mode. For example, while the machine runs a program, you can use the device manager (**[LIST PROGRAM]**) or background editor (**[EDIT]**) without stopping the program.

T2.7: List of **[EDIT]** Mode Keys and How They Operate

Name	Key	Function
Edit	[EDIT]	Lets you edit programs in the editor. You can access the Visual Programming System (VPS) and Shape Creator from the EDIT tabbed menu.
Insert	[INSERT]	Enters text from the input line or the clipboard into the program at the cursor position.
Alter	[ALTER]	Replaces the highlighted command or text with text from the input line or the clipboard.  NOTE: [ALTER] does not work for offsets.
Delete	[DELETE]	Deletes the item that the cursor is on, or deletes a selected program block.
Undo	[UNDO]	Undoes up to the last 40 edit changes, and deselects a highlighted block.  NOTE: [UNDO] does not work for deleted highlighted blocks or to recover a deleted program.

T2.8: List of **[MEMORY]** Mode Keys and How They Operate

Name	Key	Function
Memory	[MEMORY]	Selects memory mode. You run programs in this mode, and the other keys in the MEM row control the ways in which the program is run. Shows <i>OPERATION:MEM</i> in upper left display.
Single Block	[SINGLE BLOCK]	Toggles single block on or off. When single block is on, the control runs only one program block each time you press [CYCLE START] .
Graphics	[GRAPHICS]	Opens Graphics mode.

Name	Key	Function
Optional Stop	[OPTION STOP]	Toggles optional stop on or off. When optional stop is on, the machine stops when it reaches M01 commands.
Block Delete	[BLOCK DELETE]	Toggles Block Delete On or Off. When Block Delete is On, the control ignores (does not execute) the code following a Forward Slash (/), on that same line.

T2.9: List of **[MDI]** Mode Keys and How They Operate

Name	Key	Function
Manual Data Input	[MDI]	In MDI mode, you run unsaved programs or blocks of code entered from the control. Shows <i>EDIT:MDI</i> in upper left display.
Coolant	[COOLANT]	Turns the optional coolant on and off. Press [SHIFT] and then [COOLANT] to turn on the optional High-Pressure Coolant (HPC). Because HPC and regular coolant share a common orifice, you cannot activate both at the same time.
Handle Scroll	[HANDLE SCROLL]	Toggles Handle Scroll mode. This lets you use the jog handle to move the cursor in menus while the control is in jog mode.
Turret Forward	[TURRET FWD]	Rotates the tool turret forward to the next sequential tool. If Tnn is entered on the input line, the turret will advance in the forward direction to tool nn.
Turret Reverse	[TURRET REV]	Rotates the tool turret backward to the previous tool. If Tnn is entered on the input line, the turret will advance in the reverse direction to tool nn.

T2.10: List of **[HANDLE JOG]** Mode Keys and How They Operate

Name	Key	Function
Handle Jog	[HANDLE JOG]	Enters Jog mode.
.0001/.1 .001/1 .01/10 .1/100	[.0001 / .1], [.001 / 1], [.01 / 10], [.1 / 100]	Selects the increment for each click of the jog handle. When the lathe is in MM mode the first number is multiplied by ten when jogging the axis (e.g., .0001 becomes 0.001 mm). The bottom number sets speed after you press and hold an axis jog key. Shows <i>SETUP:JOG</i> in upper left display.

T2.11: List of **[ZERO RETURN]** Mode Keys and How They Operate

Name	Key	Function
Zero Return	[ZERO RETURN]	Selects Zero Return mode, which displays axis location in four different categories: Operator, Work G54, Machine, and Dist (distance) To Go. Select the tab to switch between the categories. Shows <i>SETUP: ZERO</i> in the upper-left display.
All	[ALL]	Returns all axes to machine zero. This is similar to [POWER UP] , except a tool change does not occur.
Origin	[ORIGIN]	Sets selected values to zero.
Single	[SINGLE]	Returns one axis to machine zero. Press the desired axis letter on the Alpha keyboard and then press [SINGLE] .
Home G28	[HOME G28]	Returns all axes to zero in rapid motion. [HOME G28] will also home a single axis in the same manner as [SINGLE].  CAUTION: <i>Make sure the axis motion paths are clear when you press this key. There is no warning or prompt before axis motion begins.</i>

T2.12: List of **[LIST PROGRAM]** Mode Keys and How They Operate

Name	Key	Function
List Programs	[LIST PROGRAM]	Accesses a tabbed menu to load and save programs.
Select Programs	[SELECT PROGRAM]	Makes the highlighted program the active program.
Back	[BACK ARROW] ,	Navigates to the screen you were on before the current one. This key operates like the BACK button on a web browser.

Name	Key	Function
Forward	[FORWARD ARROW] ,	Navigates to the screen you went to after the current screen, if you have used the back arrow. This key operates like the FORWARD button on a web browser.
Erase Program	[ERASE PROGRAM]	Deletes the selected program in List Program mode. Deletes the entire program in MDI mode.

Numeric Keys

Use the numeric keys to type numbers, along with some special characters (printed in yellow on the main key). Press **[SHIFT]** to enter the special characters.

T2.13: List of Numeric Keys and How They Operate

Name	Key	Function
Numbers	[0]-[9]	Types numbers.
Minus sign	[-]	Adds a minus (-) sign to the input line.
Decimal point	[.]	Adds a decimal point to the input line.
Cancel	[CANCEL]	Deletes the last character typed.
Space	[SPACE]	Adds a space to input.
Enter	[ENTER]	Answers prompts and writes input.
Special Characters	Press [SHIFT] , then a numeric key	Inserts the yellow character on the upper-left of the key. These characters are used for comments, macros, and certain special features.
+	[SHIFT] , then [-]	Inserts +
=	[SHIFT] , then [0]	Inserts =
#	[SHIFT] , then [.]	Inserts #
*	[SHIFT] , then [1]	Inserts *
‘	[SHIFT] , then [2]	Inserts `

Name	Key	Function
?	[SHIFT], then [3]	Inserts ?
%	[SHIFT], then [4]	Inserts %
\$	[SHIFT], then [5]	Inserts \$
!	[SHIFT], then [6]	Inserts !
&	[SHIFT], then [7]	Inserts &
@	[SHIFT], then [8]	Inserts @
:	[SHIFT], then [9]	Inserts :

Alpha Keys

Use the alpha keys to type the letters of the alphabet, along with some special characters (printed in yellow on the main key). Press **[SHIFT]** to enter the special characters.

T2.14: List of Alpha Keys and How They Operate

Name	Key	Function
Alphabet	[A]-[Z]	Uppercase letters are the default. Press [SHIFT] and a letter key for lowercase.
End-of-block (EOB)	[;]	This is the end-of-block character, which signifies the end of a program line.
Parentheses	[(,)]	Separate CNC program commands from user comments. They must always be entered as a pair.
Shift	[SHIFT]	Accesses additional characters on the keyboard, or shifts to lower case alpha characters. The additional characters are seen in the upper left of some of the alpha and number keys.
Special Characters	Press [SHIFT] , then an alpha key	Inserts the yellow character on the upper-left of the key. These characters are used for comments, macros, and certain special features.
Forward Slash	[SHIFT], then [;]	Inserts /

Name	Key	Function
Left Bracket	[SHIFT] , then [[	Inserts [
Right Bracket	[SHIFT] , then)]	Inserts]

Jog Keys

Name	Key	Function
Tailstock towards spindle	[TS ←]	Press and hold this key to move the tailstock towards the spindle.
Tailstock rapid	[TS RAPID]	Increases the speed of the tailstock when pressed simultaneously with one of the other tailstock keys.
Tailstock away from spindle	[TS →]	Press and hold this key to move the tailstock away from the spindle.
Axis keys	[+X/-X, +Z/-Z, +Y/-Y, +C/-C]	Press and hold an individual key or press the desired axes and use the jog handle.
Rapid	[RAPID]	Press and hold this key simultaneously with one of the above keys (X+, X-, Z+, Z-) to move that axis in the selected direction at maximum jog speed.
Chip Conveyor Forward	[CHIP FWD]	Starts optional chip conveyor in the "Forward" direction, moving chips out of the machine.
Chip Conveyor Stop	[CHIP STOP]	Stops the chip conveyor.
Chip Conveyor Reverse	[CHIP REV]	Starts the optional chip conveyor in the "Reverse" direction, which is useful in clearing jams and debris.

Override Keys

T2.15: List of Override Keys and How They Operate

Name	Key	Function
-10% Feedrate	[-10% FEEDRATE]	Decreases the current feedrate by 10%.
100% Feedrate	[100% FEEDRATE]	Sets an overridden feedrate back to the programmed feed rate.
+10% Feedrate	[+10% FEEDRATE]	Increases the current feedrate by 10%.
Handle Control Feed Rate	[HANDLE FEED]	Lets you use the jog handle to adjust the feedrate in 1% increments.
-10% Spindle	[-10% SPINDLE]	Decreases the current spindle speed by 10%.
100% Spindle	[100% SPINDLE]	Sets the overridden spindle speed back to the programmed speed.
+10% Spindle	[+10% SPINDLE]	Increases the current spindle speed by 10%.
Handle Spindle	[HANDLE SPINDLE]	Lets you use the jog handle to adjust the spindle speed in 1% increments.
Forward	[FWD]	Starts the spindle in the clockwise direction.
Stop	[STOP]	Stops the spindle.
Reverse	[REV]	Starts the spindle in the counterclockwise direction.
Rapids	[5% RAPID]/ [25% RAPID]/ [50% RAPID] / [100% RAPID]	Limits machine rapids to the value on the key.

Override Usage

Overrides let you temporarily adjust the speeds and feeds in your program. For example, you can slow down rapids while you prove out a program, or adjust the feedrate to experiment with its effect on part finish, etc.

You can use Settings 19, 20, and 21 to disable the feedrate, spindle, and rapid overrides, respectively.

[FEED HOLD] acts as an override that stops rapid and feed moves when you press it. **[FEED HOLD]** also stops tool changes and part timers, but not tapping cycles or dwell timers.

Press **[CYCLE START]** to continue after a **[FEED HOLD]**. When the Setup Mode key is unlocked, the door switch on the enclosure also has a similar result but displays *Door Hold* when the door is opened. When the door is closed, the control is in Feed Hold and **[CYCLE START]** must be pressed to continue. Door Hold and **[FEED HOLD]** do not stop any auxiliary axes.

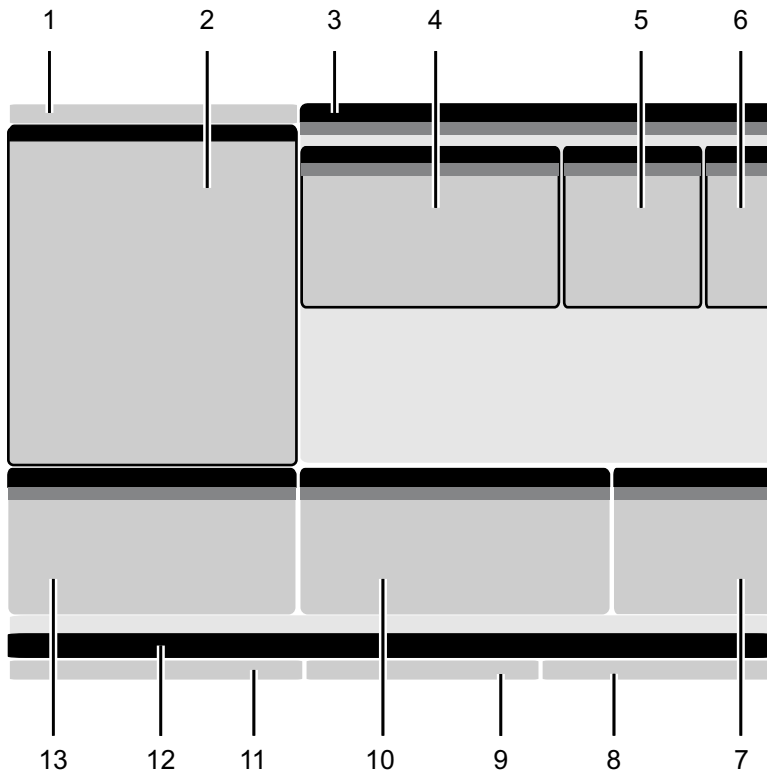
You can override the standard coolant setting by pressing **[COOLANT]**. The coolant pump remains either on or off until the next M-code or operator action (see Setting 32).

Use Settings 83, 87, and 88 to have `M30` and `M06` commands, or **[RESET]**, respectively, change overridden values back to their defaults.

2.2.4 Control Display

The control display is organized into panes that change with the different machine and display modes.

F2.9: Basic Control Display Layout in **Operation:Mem** Mode (While a Program Runs)



1. Mode, Network, and Time Status Bar
2. Program Display
3. Main Display (size varies)/Program/Offsets/Current Commands/Settings/Graphics/Editor/VPS/Help
4. Active Codes
5. Active Tool
6. Coolant

7. Timers, Counters / Tool Management
8. Alarm Status
9. System Status Bar
10. Position Display / Axis Load
11. Input Bar
12. Icon Bar
13. Spindle Status

The active pane has a white background. You can work with data in a pane only when that pane is active, and only one pane is active at a time. For example, when you select the **Tool Offsets** tab, the offsets table background turns white. You can then make changes to the data. In most cases, you change the active pane with the display keys.

Mode and Active Display Bar

The Haas control organizes machine functions into three modes: Setup, Edit, and Operation. Each mode shows on one screen all of the information you need to do tasks under that mode. For example, in Setup mode, you have access to both the work offset table, the tool offset table, and position information. Edit mode gives you access to the program editor and optional systems like Visual Programming (VPS) (which contains Wireless Intuitive Probing (WIPS)). Operation mode includes Memory (MEM), the mode in which you run programs.

F2.10: The Mode and Display bar shows [1] the current mode, [2] the network connectivity status, and [3] the time.



T2.16: Mode, Key Access, and Mode Display

Mode	Keys	Display [1]	Function
Setup	[ZERO RETURN]	SETUP: ZERO	Provides all control features for machine setup.
	[HANDLE JOG]	SETUP: JOG	
Edit	[EDIT]	ANY	Provides all program editing, management, and transfer functions.
	[MDI]	EDIT: MDI	
	[LIST PROGRAM]	ANY	
Operation	[MEMORY]	OPERATION: MEM	Provides all control features necessary to run a program.
	[EDIT]	OPERATION: MEM	Provides background editing of active programs.
	[LIST PROGRAM]	ANY	Provides background editing of programs.

Offsets Display

To access the offset tables, press **[OFFSET]** and select the **TOOL** tab or the **WORK** tab.

T2.17: Offset Tables

Name	Function
TOOL	Display and work with tool numbers and tool length geometry.
WORK	Display and work with part zero locations.

Current Commands

This section describes the Current Commands pages and the types of data they show. The information from most of these pages also appears in other modes.

Press **[CURRENT COMMANDS]** to access the tabbed menu of available Current Commands displays.

Devices -The **Mechanisms** tab on this page shows hardware devices on the machine that you can command manually. For example, you can manually extend and retract the Parts Catcher or Probe Arm. You can also manually rotate the spindle clockwise or counterclockwise at a desired RPM.

Timers Display -This page shows:

- The current date and time.
- The total power on time.
- Total cycle start time.
- Total feed time.
- M30 counters. Each time the a program reaches an **M30** command, both of these counters increment by one.
- Macro variable displays.

You also see these timers and counters in the lower-right section of the display in the **OPERATION:MEM**, **SETUP:ZERO**, and **EDIT:MDI** modes.

Macros Display -This page shows a list of the macro variables and their values. The control updates these variables as programs run. You can modify the variables in this display; refer to the Variable Display Page on page **198**.

Active Codes -This page lists the active program codes. A smaller version of this display is included on the **OPERATION:MEM** and **EDIT:MDI** mode screens. Also when you press **[PROGRAM]** in any Operation mode, you see the active program codes.

Advanced Tool Management -This page contains information the control uses to predict tool life. Here is where you create and manage tool groups, and where you enter the maximum tool load percentage expected for each tool.

For more information, refer to the Advanced Tool Management section in the Operation chapter of this manual.

Calculator -This page contains the Standard, Milling/Turning, and Tapping calculators.

Media -This page contains the **Media Player**.

Devices / Mechanisms

The **Mechanisms** page displays possible machine components and options on your machine. Select the listed mechanism using the **[UP]** and **[DOWN]** arrows for more information on its operation and use. Pages give detailed instructions on the functions of the machine components, quick tips, as well as links to other pages to help you learn about and utilize your machine.

- Select the Devices tab in the **[CURRENT COMMANDS]** menu.
- Select the Mechanisms that you want to use.

Main Spindle

F2.11: Main Spindle Device Display

08:43:48

N2

Current Commands

Devices Timers Macro Vars Active Codes ATM Calculator Media

Mechanisms

Device	State
Main Spindle	Off
Parts Catcher	Retracted
Probe Arm	Retracted

Main Spindle
Number + **F2** Set RPM
Hold **F3** **to rotate clockwise
Hold **F4** **to rotate counterclockwise

**Use [F2] to set the speed to rotate at, a value of zero will turn this feature off.
**Press and hold [F3] to rotate clockwise and [F4] to rotate counterclockwise
**Once the button is released the spindle will come to a stop

The **Main Spindle** option in **Devices** allows you to rotate the spindle clockwise or counterclockwise at a chosen RPM. The maximum RPM is limited by the machine's maximum RPM settings.

- Use the cursor arrow keys to move from field to field.
- Enter the RPM you want to rotate the spindle and press **[F2]**.
- Hold down **[F3]** to rotate the spindle clockwise. Hold down **[F4]** to rotate the spindle counterclockwise. The spindle comes to a stop when the button is released.

Parts Catcher

F2.12: Parts Catcher Device Display

08:43:56

N2

Current Commands

Devices Timers Macro Vars Active Codes ATM Calculator Media

Mechanisms

Device	State
Main Spindle	Off
Parts Catcher	Retracted
Probe Arm	Retracted

Parts Catcher

F2 Extend

****Door must be fully opened or fully closed, otherwise damage may occur.**

****Use [F2] to extend the parts catcher towards the spindle or retract for continued operation.**

The **Parts Catcher** option in **Devices** allows you to **Extend** and **Retract** the parts catcher. The door must be fully opened or fully closed.

- Use the cursor arrow keys to move from field to field.
- Press **[F2]** to extend the parts catcher and press **[F2]** to retract the parts catcher.